.

STAT 576 Empirical Process Theory and Weak Convergence Fall 2025

Course Syllabus

Lecturer: Eren C. KIZILDAĞ

Description and Intended Audience

This course is a graduate-level introduction to Empirical Process Theory and Weak Convergence, with applications geared towards modern high-dimensional statistics. We will cover central questions of Empirical Process Theory (e.g., Uniform Law of Large Numbers and Uniform Central Limit Theorem (CLT)). We will explore the CLT in depth through Lindeberg's method, and introduce the rudiments of Stein's method, a powerful tool for establishing CLTs in dependent settings and for deriving rates of convergence and finite-sample bounds.

The main goal is to train mathematically inclined PhD students in Statistics and related fields, and to equip them with the toolkit on Empirical Process Theory, Stein's method, and concentration inequalities useful for their research.

Prerequisite

Either STAT 575, 553 (or equivalent courses) or consent from the instructor.

Administrative details

Class meets: Tuesdays and Thursdays, 14:00-15:20.
Classroom: 21 Psychology Building
Canvas website: https://canvas.illinois.edu/courses/62315

Instructor: Eren C. Kizildag [kizildag@illinois.edu]
127 Computing Applications Building
Webpage: https://www.eren-kizildag.com/.

Teaching Assistants: - Rajdeep Brahma [rbrahma2@illinois.edu]

Office hours: By appointment

TA Office hours: Wednesdays 17:00 - 18:00 at 169 Computing Applications Building

Work Load

- **Grading:** Scribing (15%), Homework (60%), Final Assignment (25%).
- **Homework:** Homework assignments will be given approximately every other week. They will be submitted and graded through Gradescope; you may enroll using the link or the course entry code provided below:
  Link: https://www.gradescope.com/courses/1091646
  Entry Code: `8XBP4E`
- **Final Assignment:** By default, the final assignment will be a take-home exam (akin to a comprehensive problem set). That said, the course will develop a toolkit for modern high-dimensional statistics. Depending on our progress, I may instead convert it into an end-of-semester project; I will announce further details later.
- **Scribing:** The instructor will distribute a LaTeX template; each student is expected to scribe approximately three lectures. The scribed notes and source files should be returned to the instructor by the end of the day after the lecture.

Reading materials

- **Primary:** Lecture notes.
- **Additional books:** You do not need to purchase any textbooks; the lecture notes will be sufficient for homework and exams.

  For those interested in exploring the topics further, the following (freely available online or in the library) are recommended:

  - Martin Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge University Press, 2019.
  - Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018.
  - Lecture Notes by Sourav Chatterjee on Stein's Method: `https://souravchatterjee.su.domains/AllLectures.pdf`
  - Lecture Notes by Aditya Guntuboyina: `https://www.stat.berkeley.edu/~aditya/resources/FullNotes210BSpring2018.pdf`
  - Lecture Notes by Sasha Rakhlin: `https://www.mit.edu/~rakhlin/courses/mathstat/rakhlin_mathstat_sp22.pdf`

  Instructor's Insights: I strongly believe that Wainwright and Vershynin are essential references that any researcher in statistics, machine learning, applied probability, and theoretical computer science should be familiar with.

Lecture Topics

A list of topics to be covered is as follows.

I Stein's Method

II Concentration Inequalities: Sub-Gaussian Random Variables, Chernoff Bounds via MGF, Hoeffding's Inequality, Bernstein's Inequality, Azuma/McDiarmid's Inequality, Efron-Stein Inequality, .

III Maximal Inequalities

IV Empirical Processes: covering and packing numbers, norms of random matrices, uniform laws of large numbers, metric entropy, chaining, Gaussian techniques (Slepian, Fernique-Sudakov etc.), Rademacher complexity and symmetrization

V Applications in Modern High-Dimensional Statistics:

- *Linear Regression:* Fixed-design linear regression, basic inequality, analysis of ordinary least squares and constrained least squares estimator.
- *Sparse Recovery:* Compressed sensing, LASSO, Restricted Isometry Property
- *Statistical-Computational Trade-offs*
- *Mean-Field Models*

I reserve the right to make changes if academically advisable. Any such changes will be posted on Canvas and communicated by email.